

What we know about improving GEO, and what still blocks a definitive answer.

Three rounds are settled, one is in flight, and one structural problem stands between this benchmark and the ranked, actionable answer it exists to produce. This report states all three, with the numbers they rest on.

+0.481 CLAIM DENSITY → CITATION	0.9981 TREATMENT CEILING	0.368 API VS UI DIVERGENCE	98.9% ENGINE TRAFFIC COVERED
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01 / SETTLED

What is established

Three results have survived a full pre-registered or scaled confirmatory round. Everything actionable in this report rests on the first one.

CLAIM DENSITY RAISES CITATION — DECISIVELY

Replacing generic, unquantified phrasing with specific figures, holding document length fixed within ±3 words, raises the odds an answer engine cites the page. Corpus v0.4, 18,432 calls, individually significant for every model tested.

ENGINE MODEL	CONTROL CPR	TREATMENT CPR	DELTA
anthropic/claude-haiku-4.5	0.383	0.997	+0.614
meta-llama/llama-4-maverick	0.387	1.000	+0.613
openai/gpt-5.4-mini	0.447	0.998	+0.551
mistralai/mistral-medium-3	0.565	1.000	+0.435
deepseek/deepseek-chat	0.591	0.999	+0.408
x-ai/grok-4.3	0.615	0.993	+0.378
google/gemini-3-flash-preview	0.631	1.000	+0.369
Pooled (7 models)	0.517	0.998	+0.481

moonshotai/kimi-k2 excluded from pooled figures under the standing data-health rule. Not pre-registered before collection, so recorded as strong confirmatory evidence rather than a citable round.

KEYWORD STUFFING DOES NOT DECREASE CITATION — IT SLIGHTLY RAISES IT

The one fully pre-registered round. Keywords repeated 4–8 times into content that answers the question no better, length-matched, no new facts. **H6 falsified:** pooled delta +0.038, 95% CI [+0.004, +0.075], $p = 0.045$ across 7 models. A small effect dwarfed by prompt and domain variance — but consistently positive, not one outlier. A follow-up found it wins citations *without* measurably lower claim fidelity, confirmed by two independent judges at 93.2% agreement.

API-PLANE RESULTS MAY NOT TRANSFER TO WHAT USERS SEE

Divergence between the API-with-search plane and a logged-out consumer UI: 0.368, 95% CI [0.139, 0.625], driven mostly by the API returning zero citations on 3 of 12 prompts where the consumer UI cited real sources. Nobody had published a magnitude for this before. It bounds how far any Tier 1 result can be claimed to describe real usage.

Why the corpus cannot rank tactics

THE CEILING

The v0.4 treatment condition sits at **CPR 0.9981**. Headroom remaining: **0.0019**.

That number is the single thing standing between this project and a practitioner-facing answer. The corpus proves *specific facts beat no facts* — decisively, across every engine. But statistics, attributed quotations, source citations, answer-first structure and FAQ blocks would all land at 1.000 and be indistinguishable. You cannot rank what you cannot separate.

TARGET FORMAT	TREATMENT CPR	HEADROOM
docs · news · product	1.0000	0.0000
blog	0.9985	0.0015
reference	0.9963	0.0037
forum	0.9940	0.0060

This is the same wall kwstuff-v1 hit at 0.985 and kwstuff-v2 hit at 0.911–0.984. It is structural, not accidental: the control is engineered to be fully orthogonal and the treatment to answer the question outright. Maximum contrast is ideal for an existence proof and useless for a ranking.

The obvious fix is already falsified. kwstuff-v2 tested a half-specific baseline on the theory that partial completeness would land mid-range; it saturated at the same ceiling. **These models discriminate on presence versus absence of any specific on-topic fact, not on how complete the answer is.**

Make the documents compete

In the current design only the target answers the question, so citation measures presence. If all six documents answer it, citation measures preference — which is what ranking tactics requires.

The evidence that this regime exists and is currently being wasted is already in the v0.4 data. Engines do not cite one source; they select several, and they narrow when one document answers well.

2.72

DOCS CITED PER ANSWER, OF 6

3.17

DOCS CITED — CONTROL

2.27

DOCS CITED — TREATMENT

That narrowing is the engine discriminating. The present design simply gives it nothing to discriminate between. A corpus where every document answers puts each one near 0.4–0.6 with headroom both ways, and tactics can finally separate.

THE ROUND THAT WOULD PRODUCE THE ANSWER

Six documents all answering the question. The target varies by tactic — statistics, attributed quotations, source citations, answer-first structure, FAQ — each length-matched against a plain-but-complete baseline, across the five engines weighted by share. Output: a ranked table reading *“for ChatGPT, do X; it is worth N points; here is the interval.”*

The hard part is engineering six documents that all answer without saturating. The kwstuff-v2 failure is the map of where that goes wrong, and the reason a pre-committed real-model spot check is now standing practice before any full round.

The length-only round (H7)

Every corpus version in this project is length-matched within ± 3 words because length is assumed to be a confound. That assumption had never been measured. This round measures it.

H7: adding words without adding facts does not increase citation. A dose ladder — the same v0.4 control text padded to 1.31×, 1.58× and 2.06× with discourse filler that states no fact, names no entity, contains no numeral, and shares no content word with its own prompt. 5 engines × 48 prompts × 4 conditions × 24 runs = 23,040 calls.

It is a **negative control**. If padding does nothing, it hardens every other result in the repo and validates CPR against the confound that inflated the widely-quoted Princeton figures. It will not, by itself, tell anyone what to do.

SPOT CHECK: THE GATE FAILED, AND THE FAULT WAS IN THE GATE

2,304 calls, 0 errors, \$1.98. The pre-registered gate asked that control CPR reproduce v0.4's 0.38–0.60 band; only 2 of 8 models did. The diagnosis cleared the corpus: 0.503 is v0.4's control pooled over **48** prompts, the spot check runs **6**, and per-prompt control CPR spans 0.219–0.896.

COMPARISON	CONTROL CPR	N
v0.4 control, all 48 prompts	0.503	9,216
v0.4 control, these 6 prompts	0.646	1,152
Length-only control, these 6 prompts	0.622	576

Delta -0.024 , on text that is byte-identical by construction. The gate was an operationalisation error — mismatched denominators — and was restated in subset-matched form, with the whole sequence logged as a deviation.

A FINDING INDEPENDENT OF THE ROUND

`mistralai/mistral-medium-3` now fails data health at **17.7%** no-cite, and its control CPR fell $0.590 \rightarrow 0.222$ versus v0.4 on identical text. That is an instruction-following regression in the model since v0.4 ran, not a corpus effect. v0.4's published round did not exclude it; future rounds should. `kimik2` sits at 20.5%, consistent with its known 34.4%.

Engines, ranked by market share

This benchmark exists to tell someone what to change. Their traffic is not split evenly across engines, so pooled figures are share-weighted and per-engine results stay primary.

ENGINE	SHARE	MODEL	POOL WEIGHT
ChatGPT (incl. Copilot)	54.7%	openai/gpt-5.4-mini	55.9%
Gemini	27.8%	google/gemini-3-flash-preview	28.4%
Claude	9.2%	anthropic/claude-haiku-4.5	9.4%
DeepSeek	3.6%	deepseek/deepseek-chat	3.7%
Grok	2.5%	x-ai/grok-4.3	2.6%
Perplexity	1.1%	not measurable in Tier 1	—

Similarweb Gen AI worldwide traffic share, August 2026. Panel covers 98.9% of measured gen-AI traffic. These are vendor-panel numbers inside a project that rejects vendor-panel numbers; they live in one file with their source and date so they can be replaced in a single edit, and the per-engine table never depends on them.

PERPLEXITY CANNOT BE TESTED THIS WAY

Given two numbered sources and instructed to use only those, perplexity/sonar returned citations [2] [4] [9] [10] [12] [15] — indices from its own live web results. It breaks the Tier 1 constraint that retrieval is held constant, and the failure is **silent**: the parser bounds-checks, so Sonar records as a near-total no-cite model rather than an incompatible one. The most GEO-targeted engine in existence is the one this harness structurally cannot measure.

What a practitioner can do today

Drawn only from what is settled. Everything below rests on the claim-density result; the ranking among tactics is what the competitive-corpus round would establish.

- 1 **Put a specific, checkable figure on every substantive claim.** This is the one intervention with a large, causally-tested effect across every engine measured — $0.517 \rightarrow 0.998$ CPR. Replace “significantly faster” with “12 ms median latency, measured January 2026”. Do it without lengthening the page; the effect was measured at fixed length.
- 2 **Do not fear keyword repetition, and do not rely on it.** Stuffing measured $+0.038$ — real, positive, and small enough to be dwarfed by prompt and domain variance. It is not a lever worth optimising.
- 3 **Do not judge yourself on word-share metrics.** An independent replication measured a significant +29% from meaningless filler alone on a position-adjusted word-count metric. Track whether you were cited at all and what share of citations you won.
- 4 **Treat single-engine results as single-engine.** The same replication found effects roughly halving between a cheap engine and a stronger one, with only claim density surviving both.
- 5 **Remember retrieval.** Everything here assumes your page is already in the candidate set. Tier 1 holds retrieval constant by construction and can say nothing about getting there.

What none of this shows

Retrieval is held constant. These rounds measure the synthesis stage only — which of the pages already in front of an engine it leans on. Crawlability, indexation and freshness act earlier and are not tested.

Transfer is unquantified beyond one study. The 0.368 API-versus-UI divergence is the only measurement of the gap, on one vendor, at small n .

Only the target is optimised. Results describe the returns to being the only page in a result set doing GEO. If every competitor optimises, the advantage should compress.

The corpus is synthetic and frozen. Realistic and varied, and the engines discriminate on it sharply, but it is not a sample of the live web.